LAB EXERCISE 20

App Deployment Using ECS and ECR

Name: Mohd Anas

SAP ID:500121888

Roll No:R2142230512

Course: B.Tech

Objective

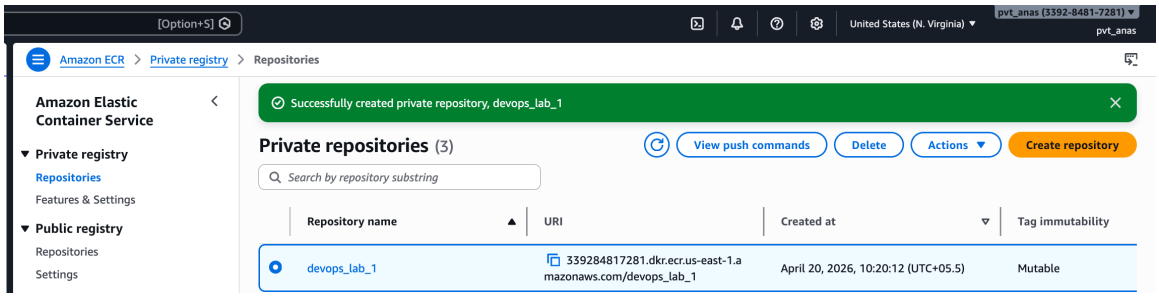
To deploy a containerized web application using Amazon ECR and ECS with Fargate.

Prerequisites

AWS Account, Docker Installed, AWS CLI Configured, Basic knowledge of containers.

Procedure

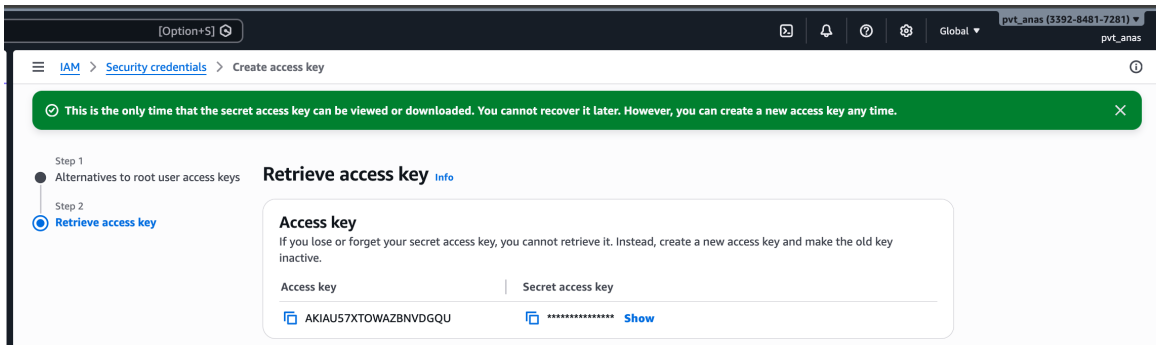
Step 1: Create ECR Repository



Step 2: Configure AWS CLI

Command:

```
aws configure
```




```
mohdanas@Mohds-MacBook-Air ~ % aws configure
AWS Access Key ID [****************IDUG]: AKIAU57XTOWAZBNVDGQU
AWS Secret Access Key [****************mbyO]: 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
Default region name [eu-west-1]:
Default output format [json]:
```

Step 3: Authenticate Docker to ECR

Command:

```
aws ecr get-login-password --region us-east-1 | docker login --username
AWS --password-stdin <account-id>.dkr.ecr.us-east-1.amazonaws.com
```

```
mohdanas@Mohds-MacBook-Air ~ % aws ecr get-login-password --region us-east-1 | docker login --username
AWS --password-stdin 339284817281.dkr.ecr.us-east-1.amazonaws.com
Login Succeeded
```

Step 4: Build Docker Image

Command:

```
docker build -t devops-repo .
```

Step 5: Tag Docker Image

Command:

```
mohdanas@Mohds-MacBook-Air my-web-app % docker build -t devops_lab_1 .
[+] Building 2.3s (9/9) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                  0.0s
 => => transferring dockerfile: 102B                                                  0.0s
 => [internal] load metadata for docker.io/library/httpd:latest                       2.2s
 => [auth] library/httpd:pull token for registry-1.docker.io                          0.0s
 => [internal] load .dockerignore                                                     0.0s
 => => transferring context: 2B                                                       0.0s
 => [1/3] FROM docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b4  0.0s
 => => resolve docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b4  0.0s
 => [internal] load build context                                                     0.0s
 => => transferring context: 32B                                                      0.0s
 => CACHED [2/3] WORKDIR /usr/local/apache2/htdocs/                                   0.0s
 => CACHED [3/3] COPY index.html .                                                    0.0s
 => exporting to image                                                                0.0s
 => => exporting layers                                                               0.0s
 => => exporting manifest sha256:6c5f9ce86e2e081f36bfefe74291bf88613c1725c2b8d8ca23fb76dc17c98a4  0.0s
 => => exporting config sha256:245e3087d5173c3994bc9e9b32f7975422529dc8472d5d762d1cc3a7bf45388e  0.0s
 => => exporting attestation manifest sha256:a7ad54bac9e8536f747953c96a25133f95c26adf6733ceb4a68  0.0s
 => => exporting manifest list sha256:424688dcbac1cb2f6676deef6f20c6c91779e71b118f1da783bd66e9a4  0.0s
 => => naming to docker.io/library/devops_lab_1:latest                                0.0s
 => => unpacking to docker.io/library/devops_lab_1:latest                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/sw8vxdn9m35b9ivdrc91ad
pxp
```

`docker tag devops-repo:latest <account-id>.dkr.ecr.us-east-1.amazonaws.com/devops-repo:latest`

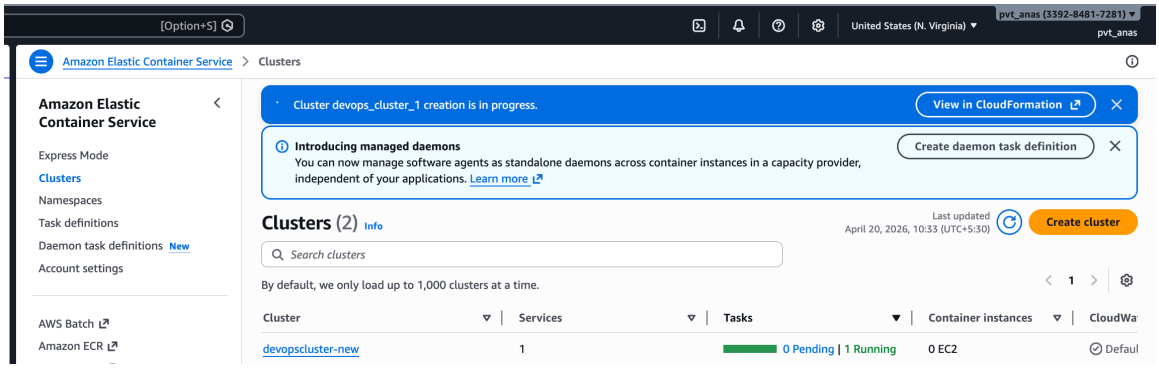
```
mohdanas@Mohds-MacBook-Air my-web-app % docker tag devops_lab_1:latest 339284817281.dkr.ecr.us-east-1.a
mazonaws.com/devops_lab_1:latest
mohdanas@Mohds-MacBook-Air my-web-app %
```

Step 6: Push Docker Image

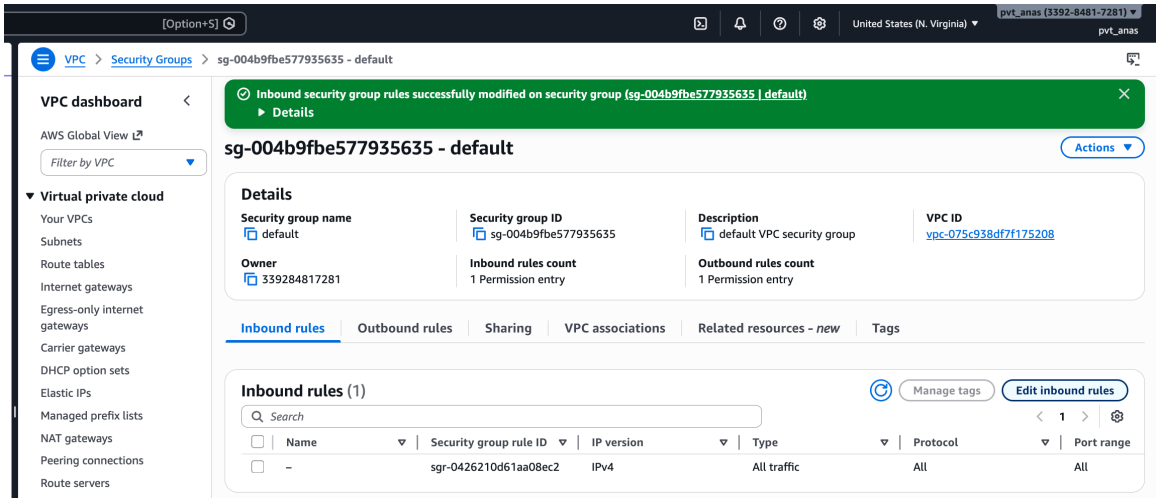
Command:

`docker push <account-id>.`dkr.ecr.us-east-1.amazonaws.com/devops-repo:latest

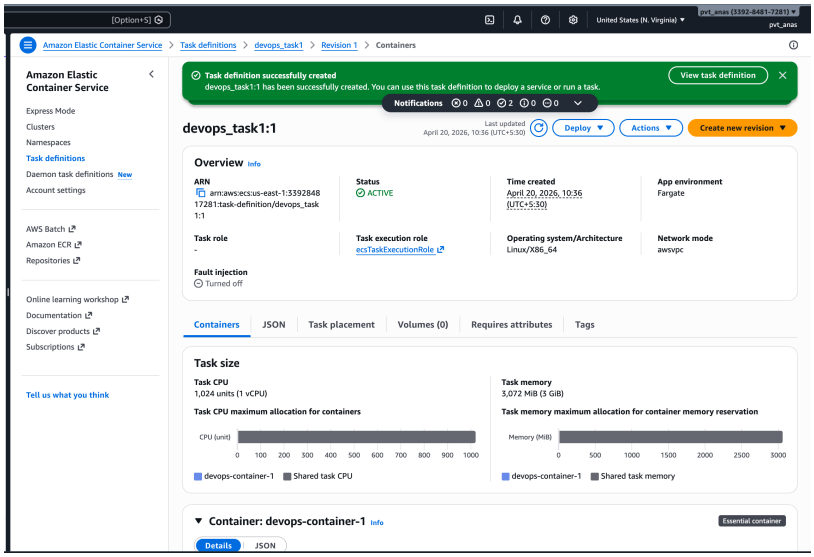
```
mohdanas@Mohds-MacBook-Air my-web-app % docker tag devops_lab_1:latest 339284817281.dkr.ecr.us-east-1.a
mazonaws.com/devops_lab_1:latest
mohdanas@Mohds-MacBook-Air my-web-app %
```



Step 7: Create ECS Cluster



Step 8: Create Task Definition

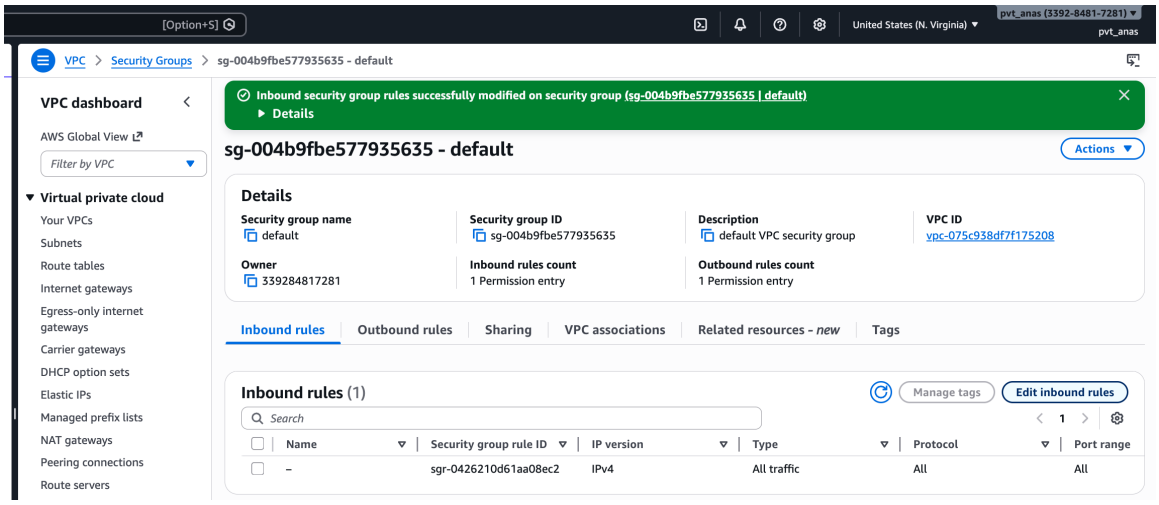


Step 9: Create Service

Step 10: Configure Security Group

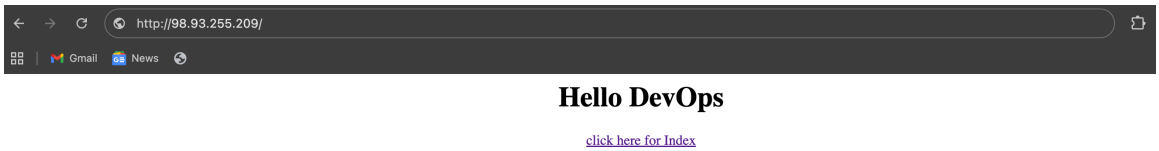
Command:

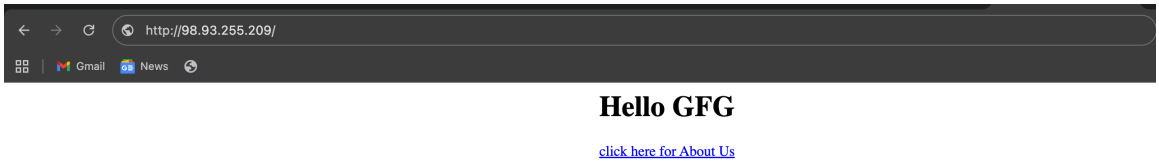
```
Allow HTTP Port 80 from 0.0.0.0/0
```



Step 11: Verify Deployment

Command:




http://98.93.255.209/
Gmail
News
Hello GFG
click here for About Us